Supplementary Material: Anchor the Head: Alignment Drift in Open-Vocabulary Segmentation

WACV 2027, Paper 3458

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This supplement records reproducibility details for the experiments reported in the main paper. The main paper is self-contained. The material here clarifies how the released JSON files, split files, and scripts support staged reproduction. The public repository contains code, splits and result records.

Evaluation protocol per table

Different comparisons use different image sets and resolutions. Table 1 lists the main-paper protocols, and Table S0 lists the supplementary protocols. These summaries identify the evaluation support for comparisons across tables.

Table S0: Supplementary evaluation protocols. Masked crops use 256×256 scoring and dense CLIP uses patch resolution. Main-paper protocols appear in Table 1.

Table	Stack and data	Images	Splits
S1, S5	ADE20K masked-crop	347	5
S3	ADE20K masked-crop	347	seed 0
S2, S4	ADE20K masked-crop	cacheable subset	as Table 2
S6, S7, S8	ADE20K dense-CLIP	2000	5
S10	Dense-CLIP timing	GPU: 2000; CPU: 100	1 (training)
S9	COCO fit, ADE dense eval	600 fit, 2000 eval	whole ADE vocabulary

A Reproducibility

Model inference and head training use a single GPU and fine-tune neither frozen model. We use a frozen SAM mask proposer and a frozen CLIP vision–language model and evaluate on ADE20K-150 validation. The masked-crop pipeline can only score images for which SAM emits at least two proposals, so the headline accuracy tables evaluate the SAM-cacheable subset of the 2000-image validation split rather than all 2000 images (the independent rebuild cached 1623 images; see `config.n_val` in `results/rebuild_mc/metrics_calib.json`). The original cache behind Table 2 and Tables S2 and S4 has no recorded image count. Calibration and encoder comparisons use 347 validation images over five splits and 300 training images; the edge ablation uses seed 0 only. Dense-CLIP scores all 2000 validation images at patch resolution. Each ADE split holds out 30 of 150 classes. Masked-crop splits use `default_rng(seed)` for seeds 0–4; dense splits use five successive permutations from one `RandomState(0)` stream. Means and standard deviations are over splits where indicated; bootstrap intervals resample images at seed 0. All methods start from the *identical* frozen-CLIP per-mask logits, so each comparison isolates the reasoning or calibration layer from the score. Post-hoc calibrators (global temperature, per-class temperature, vector scaling) are fit only on seen-class training masks, exactly the supervision a deployed open-vocabulary system would have. The

proposal graph is built from mask IoU, boundary overlap, and positive CLIP-region cosine similarity, and diffusion mixes the frozen CLIP logits back at every step, a class-symmetric and parameter-free operation. Reproduction is staged by cost in `REPRODUCE.md`. JSON-only generators rebuild Tables 2, 3, 5, 6, S2, S4 and S6–S10, with manuscript-specific captions and layout. The S1/S3 writers require a feature cache, and Table 4 and S5 are transcribed from records. Masked-crop calibration runs with CPU `numpy`; dense-CLIP and VOC evaluation require CUDA and separate dense caches (`MODELS.md`). Five masked-crop ADE files (`ade20k_seed*`), five dense ADE files (`ade20k_dense_seed*`), and five VOC files (`voc_seed*`) are provided under `splits/` with checksums.

Additional results

The result records and generation scripts are described in `scripts/EXTENDED.md` in the public repository.

B Full calibration study

We compare calibrators on identical mask logits in Table S1. Global temperature gives the lowest overall ECE, while diffusion and global temperature have similar novel-class IoU. Figure S1 separates overall and novel-class ECE. Per-class temperature and vector scaling improve the aggregate score while increasing novel-class error, showing how calibration averaged over the data can hide a failure on held-out classes.

Table S1: Calibration on identical frozen-CLIP mask logits for 347 ADE20K validation images over five splits.

Method	mIoU	novel-IoU	ECE↓	novel ECE↓	seen ECE↓
Uncalibrated baseline	$8.9_{\pm 0.0}$	$8.4_{\pm 0.9}$	$14.6_{\pm 0.0}$	$12.8_{\pm 3.0}$	$15.3_{\pm 0.8}$
Temperature scaling (global T)	$8.8_{\pm 0.0}$	$8.5_{\pm 0.8}$	$3.5_{\pm 0.6}$	$6.6_{\pm 3.1}$	$3.1_{\pm 0.8}$
Per-class temperature	$3.6_{\pm 0.6}$	$0.5_{\pm 0.6}$	$12.6_{\pm 2.8}$	$27.9_{\pm 3.5}$	$8.9_{\pm 2.5}$
Vector/Platt scaling	$3.7_{\pm 0.6}$	$0.0_{\pm 0.0}$	$16.2_{\pm 0.7}$	$34.0_{\pm 4.4}$	$12.0_{\pm 1.6}$
Graph diffusion	$9.1_{\pm 0.0}$	$8.5_{\pm 1.0}$	$5.9_{\pm 0.0}$	$5.2_{\pm 1.7}$	$6.6_{\pm 0.9}$

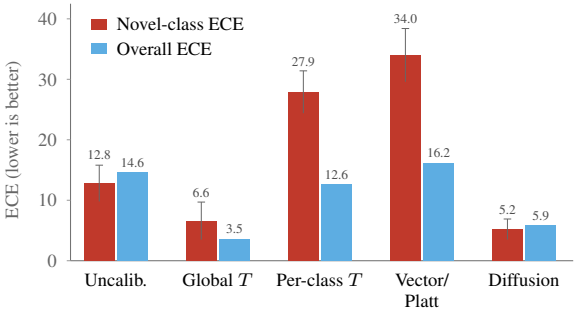


Figure S1: Overall (blue) and novel-class (red) ECE on ADE20K masked crops. Values follow Table S1. Lower is better.

C Diffusion sweep and edge ablation

We vary the diffusion weight α and iteration count k in Table S2. Lower ECE does not imply higher mIoU. At α=0.6, k=3, ECE reaches 2.2 while mIoU is 8.5, below the 8.9 baseline. Table S3 isolates the evidence used to connect masks. Boundary and CLIP-similarity edges retain novel-class IoU, while IoU-only edges give 9.57 against the seed-0 baseline of 9.75. Thus geometric overlap alone does not reproduce the behavior of the full graph.

Table S2: Diffusion weight (α) and step (k) ablation on the original ADE20K masked-crop cache. α=0 denotes independent scoring.

Config	mIoU	hIoU	novel-IoU	ECE↓
baseline (α=0)	8.9	9.1	9.4	14.6
α=0.2, k=1	8.8	9.0	9.4	9.3
α=0.2, k=2	8.7	8.9	9.2	9.8
α=0.2, k=3	8.7	8.9	9.2	9.7
α=0.4, k=1	8.6	9.1	10.1	4.5
α=0.4, k=2	**9.0**	**9.4**	10.0	5.9
α=0.4, k=3	8.9	9.3	10.0	5.4
α=0.6, k=1	8.1	8.6	9.8	2.4
α=0.6, k=2	8.9	9.2	9.8	3.5
α=0.6, k=3	8.5	9.1	**10.5**	**2.2**
α=0.8, k=1	7.0	7.5	8.7	4.6
α=0.8, k=2	**9.0**	9.3	10.0	4.5
α=0.8, k=3	7.7	8.1	9.0	3.0

Table S3: Diffusion edge-source ablation on 347 ADE20K validation images (α=0.4, k=2, seed 0).

Edge source	mIoU	novel-IoU	ECE↓	novel ECE↓
all edges	9.1	10.2	5.9	5.5
IoU only	8.7	9.6	6.4	5.9
boundary only	9.0	10.5	5.9	5.3
CLIP-cosine only	9.1	10.2	5.8	5.4

D Full decoder ablation

We remove one decoder component at a time in Table S4. A minus sign denotes removal, and MLP denotes a multilayer perceptron applied independently to each node. The row labels abbreviate calibration as calib and prototype refinement as proto. Removing message passing raises seen-class IoU to 9.8, yet novel-class IoU remains 1.5 against the baseline's 9.4. Removing prototype refinement reduces both. The failure therefore persists across decoder variants, even when seen-class performance improves.

Table S4: Graph, calibration and prototype-refinement ablations of the trained decoder on the original ADE20K masked-crop cache.

Variant	mIoU	seen-IoU	novel-IoU	hIoU	ECE↓
Proposal+CLIP baseline	8.9	8.7	9.4	9.1	14.6
full (graph+residual+calib+proto)	7.1	8.6	1.1	2.0	23.8
− graph (per-node MLP)	8.1	9.8	1.5	2.7	24.6
− calibration head	7.1	8.6	1.3	2.2	23.6
− prototype refinement	4.3	5.5	0.0	0.0	27.8

E Stronger-encoder probe

We test whether the crop representation changes the comparison in Table S5. The standard bounding-box (bbox) crop is labeled "weak," and the context-padded crop "strong." Slash-separated entries report global temperature followed by diffusion. Context increases baseline novel-class IoU from 8.4 to 9.3, while global temperature remains

the better overall calibrator. Improving the base representation and calibrating its outputs address different aspects of performance.

Table S5: Standard and context-padded crops (padding 0.25) with calibration on ADE20K.

Encoder	Method	mIoU	novel-IoU	ECE↓	novel ECE↓
bbox crop (weak)	baseline	8.9	8.4	14.6	12.8
	global T / diffusion	8.8 / 9.1	8.5 / 8.5	3.5 / 5.9	6.6 / **5.2**
context crop (strong)	baseline	9.2	9.3	16.0	13.8
	global T / diffusion	9.3 / 9.3	9.4 / 9.3	3.8 / 6.9	8.2 / **7.7**

F PASCAL VOC regime

Table 4 uses frozen CLIP ViT-B/16 with Correlative Self-Attention on VOC2012 validation, at patch resolution (75.3 mIoU; the recorded full-resolution base is 81.8). Six of the 20 object classes are novel in each of five sequential permutations from `RandomState(0)`. Background and void labels are excluded. Corrections are fitted on seen-class patches from the training split. The released `results/metrics_voc.json` gives split variation: novel-class IoU is 75.2 ± 8.0 for the frozen base and global temperature, 0.0 ± 0.0 for per-class temperature, 77.1 ± 8.1 for diffusion, and 16.1 ± 5.5 for the decoder. Global-temperature ECE is 4.3 ± 0.4.

G Calibration, retention and selective risk

We report area under the risk–coverage curve (AURC) alongside IoU and ECE in Table S6. Risk is prediction error among patches retained by confidence, and coverage is their fraction of the evaluated set. TR denotes a logit trust region. The trained corrections obtain lower AURC while losing novel-class IoU, showing that confidence ranking alone does not establish novel-class retention. A dash denotes an unreported value.

Table S6: Accuracy, calibration and selective-prediction risk for ADE20K dense-CLIP corrections over five splits.

Method	mIoU	hIoU	ECE↓	novel-IoU	AURC↓
Graph diffusion (anchored)	15.0	15.2	27.8	15.7	49.2
Graph Laplacian (generic, no anchor)	15.5	15.8	28.5	16.3	48.6
Trained MLP (feature input)	5.5	0.2	2.6	0.1	38.2
Trained decoder (logit-space TR)	10.5	7.6	11.0	5.8	34.2
Text-conditioned per-class calibrator	8.0	3.2	19.0	1.9	–

H Frontier grid

We list the output-gate and residual-scale sweeps behind Figure 3 in Table S7. Increasing g or η moves toward the unanchored head and progressively reduces novel-class IoU. At g=0.1, seen-class IoU exceeds the frozen baseline while novel-class IoU remains close to it. The useful operating region is therefore near the frozen predictor, with a measurable adaptation–retention trade-off.

Table S7: Anchoring sweeps on ADE20K dense CLIP over five splits. g mixes probabilities and η scales logit residuals, from frozen (0) to unanchored (1). C denotes novel-class IoU retention.

Setting	mIoU	seen-IoU	novel-IoU	hIoU	ECE↓	C
frozen baseline	$14.5_{\pm 0.0}$	$14.3_{\pm 0.3}$	$15.1_{\pm 1.0}$	$14.7_{\pm 0.4}$	$27.2_{\pm 0.0}$	1.00
B3 output gate g=0.1	$15.9_{\pm 0.3}$	$16.2_{\pm 0.3}$	$14.7_{\pm 1.4}$	$15.4_{\pm 0.8}$	$36.5_{\pm 1.6}$	0.97
B3 output gate g=0.2	$15.1_{\pm 0.4}$	$15.7_{\pm 0.4}$	$13.0_{\pm 2.1}$	$14.1_{\pm 1.3}$	$32.8_{\pm 2.1}$	0.86
B3 output gate g=0.4	$12.4_{\pm 0.5}$	$13.6_{\pm 0.4}$	$8.0_{\pm 2.5}$	$9.8_{\pm 2.0}$	$23.3_{\pm 2.3}$	0.53
B3 output gate g=0.6	$9.6_{\pm 0.5}$	$11.0_{\pm 0.4}$	$3.8_{\pm 2.1}$	$5.3_{\pm 2.5}$	$14.0_{\pm 2.3}$	0.25
B3 output gate g=0.8	$7.2_{\pm 0.3}$	$8.9_{\pm 0.3}$	$0.6_{\pm 0.6}$	$1.1_{\pm 1.0}$	$5.3_{\pm 2.0}$	0.04
B3 output gate g=1.0	$6.5_{\pm 0.3}$	$8.1_{\pm 0.3}$	$0.1_{\pm 0.1}$	$0.1_{\pm 0.3}$	$3.9_{\pm 2.5}$	0.01
B4 residual scale η=0.0	$14.5_{\pm 0.0}$	$14.3_{\pm 0.3}$	$15.1_{\pm 1.0}$	$14.7_{\pm 0.4}$	$27.2_{\pm 0.0}$	1.00
B4 residual scale η=0.25	$14.2_{\pm 0.4}$	$15.5_{\pm 0.5}$	$9.3_{\pm 2.3}$	$11.4_{\pm 1.8}$	$29.7_{\pm 2.7}$	0.61
B4 residual scale η=0.5	$11.2_{\pm 0.3}$	$13.1_{\pm 0.5}$	$3.5_{\pm 1.9}$	$5.3_{\pm 2.5}$	$19.2_{\pm 1.9}$	0.23
B4 residual scale η=0.75	$8.1_{\pm 0.3}$	$10.0_{\pm 0.4}$	$0.5_{\pm 0.5}$	$0.8_{\pm 0.8}$	$7.3_{\pm 2.1}$	0.03
B4 residual scale η=1.0	$6.5_{\pm 0.3}$	$8.1_{\pm 0.3}$	$0.1_{\pm 0.1}$	$0.1_{\pm 0.3}$	$3.9_{\pm 2.5}$	0.01

I Confound ablation

Table S8 tests one alternative explanation per row. Class-balanced sampling equalizes training-class representation. Focal loss downweights easy examples, while the cosine-margin variant subtracts a fixed margin from the target-class logit. Pure patches have at least 75% label agreement in their 3×3 patch neighborhood. These changes leave novel-class IoU low. The 30-class supervision setting is a partial exception, but its standard deviation (4.3) exceeds its mean (3.3). Tightening the trust region provides the consistent recovery in this comparison.

Table S8: Single-factor head ablations on ADE20K dense CLIP over five splits. C4 varies the trust-region bound τ. C denotes novel-class IoU retention, and drift follows Eq. (2).

Condition	mIoU	seen-IoU	novel-IoU	hIoU	ECE↓	drift	C
Uncalibrated (frozen baseline)	$14.5_{\pm 0.0}$	$14.3_{\pm 0.3}$	$15.1_{\pm 1.0}$	$14.7_{\pm 0.4}$	$27.2_{\pm 0.0}$	0.0	1.00
Seen-only head (CE)	$6.5_{\pm 0.4}$	$8.2_{\pm 0.5}$	$0.1_{\pm 0.2}$	$0.2_{\pm 0.4}$	$3.9_{\pm 2.7}$	56.9	0.01
C1 class-balanced CE	$9.6_{\pm 0.5}$	$12.0_{\pm 0.6}$	$0.0_{\pm 0.0}$	$0.0_{\pm 0.1}$	$16.5_{\pm 3.0}$	36.3	0.00
C2 focal loss	$7.0_{\pm 0.5}$	$8.7_{\pm 0.5}$	$0.1_{\pm 0.1}$	$0.1_{\pm 0.3}$	$4.5_{\pm 1.5}$	54.7	0.01
C2 cosine-margin loss	$4.9_{\pm 0.3}$	$6.0_{\pm 0.4}$	$0.1_{\pm 0.1}$	$0.1_{\pm 0.3}$	$20.3_{\pm 3.7}$	63.0	0.01
C3 learned logit scale	$7.8_{\pm 0.4}$	$9.7_{\pm 0.5}$	$0.3_{\pm 0.4}$	$0.6_{\pm 0.7}$	$5.2_{\pm 2.4}$	55.5	0.02
C4 trust region τ=0.5	$15.2_{\pm 0.4}$	$15.4_{\pm 0.5}$	$14.6_{\pm 0.9}$	$14.9_{\pm 0.5}$	$31.2_{\pm 1.9}$	5.6	0.96
C4 trust region τ=1.0	$14.3_{\pm 0.3}$	$14.8_{\pm 0.3}$	$12.1_{\pm 2.1}$	$13.2_{\pm 1.2}$	$28.2_{\pm 2.0}$	19.4	0.80
C4 trust region τ=2.0	$10.6_{\pm 0.4}$	$11.7_{\pm 0.4}$	$5.9_{\pm 2.1}$	$7.6_{\pm 1.9}$	$10.9_{\pm 1.6}$	44.6	0.39
C4 trust region τ=4.0	$6.8_{\pm 0.4}$	$8.3_{\pm 0.5}$	$0.5_{\pm 0.6}$	$0.9_{\pm 1.0}$	$3.3_{\pm 2.4}$	55.8	0.03
C5 supervision breadth: 30 seen classes	$3.7_{\pm 2.6}$	$3.8_{\pm 2.2}$	$3.3_{\pm 4.3}$	$3.1_{\pm 3.5}$	$22.2_{\pm 12.5}$	63.7	0.22
C5 supervision breadth: 60 seen classes	$2.7_{\pm 0.2}$	$3.3_{\pm 0.3}$	$0.2_{\pm 0.3}$	$0.4_{\pm 0.5}$	$19.0_{\pm 6.5}$	67.3	0.01
C5 supervision breadth: 90 seen classes	$4.3_{\pm 0.5}$	$5.4_{\pm 0.6}$	$0.1_{\pm 0.1}$	$0.1_{\pm 0.2}$	$12.8_{\pm 3.8}$	62.7	0.00
C5 supervision breadth: 120 seen classes	$6.5_{\pm 0.4}$	$8.2_{\pm 0.5}$	$0.1_{\pm 0.2}$	$0.2_{\pm 0.4}$	$3.9_{\pm 2.7}$	56.9	0.01
C6 spatially pure patches only	$4.8_{\pm 0.1}$	$5.9_{\pm 0.2}$	$0.2_{\pm 0.2}$	$0.4_{\pm 0.4}$	$4.0_{\pm 2.2}$	57.5	0.01

J COCO-Stuff transfer

We fit on COCO-Stuff and evaluate on ADE20K without fitting to ADE labels (Table S9). The acc column is patch accuracy. Class-indexed parameters need a vocabulary mapping before transfer, whereas text-conditioned and class-agnostic corrections apply directly. The embedding penalty weight λ controls cosine anchoring. Its improvement from 14.9 to 15.6 mIoU shows that preserving the frozen representation can also benefit a correction trained on another dataset.

Table S9: Cross-vocabulary transfer from 600 COCO-Stuff-171 images to all 2000 ADE20K validation images using the same frozen backbone.

Correction (fit on COCO-Stuff)	mIoU	acc	ECE↓	drift
None (frozen baseline)	14.5	35.4	27.2	0.0
Global temperature	14.5	35.4	8.4	13.7
Text-conditioned calibrator	14.3	43.1	6.5	33.8
Class-agnostic emb. head (λ=0)	14.9	56.4	4.6	46.1
Class-agnostic emb. head (λ=1)	15.6	55.8	8.9	41.6
Diffusion (fit-free)	15.0	35.9	27.8	0.2
Per-class T / logit-space head: *not applicable* (parameters indexed by COCO classes)				

K Cost

We use one H100 Multi-Instance GPU (MIG) 2g.20GB partition. Diffusion on cached features takes 0.9 ms per image, and our dense MLP trains in 1.6 s per split. Table S10 reports the cost of correcting cached features, excluding backbone feature extraction. We measure wall-clock time, time per image and peak GPU memory. The sub-millisecond GPU diffusion cost makes correction inexpensive once features are available. Timing includes graph construction and device transfers. GPU diffusion evaluates 2000 images and CPU diffusion 100. Training uses 341 004 patches from one split. Graphs average 576 nodes and 4608 directed neighbor entries per image (k_{nn}=8). GPU memory in the CPU row is resident allocation.

Table S10: Correction runtime and memory on dense CLIP using one H100 MIG 2g.20GB slice.

Component	wall clock	per image	peak GPU mem
Graph diffusion (GPU, full val)	1.8 s	0.9 ms	259 MB
Graph diffusion (CPU, 100 imgs)	1.4 s	14 ms	242 MB
Global temperature fit	1.5 s	–	1475 MB
Decoder training (12 ep, 1 split)	1.6 s	–	346 MB
Decoder parameters: 169 878			

L Full specification of the audited corrections

Dense residual heads. Our dense-CLIP heads use a two-layer multilayer perceptron (MLP) $\mathbb{R}^{512} \to 256 \to \mathbb{R}^{150}$ with rectified linear units (ReLU), applied to patch features. Frozen logits have scale 40. Tables 3 and S6 use a 0.5 output multiplier for the feature-input MLP; the Table 5 alignment family adds the unscaled logit residual. B2 instead predicts a 512-dimensional embedding residual. Training uses Adam, learning rate 10^{-3}, batches of 8192 patches and 12 epochs, with cross-entropy on seen-class patches from the 600-image train cache. No novel labels are used for stopping or selection. Frozen-logit distillation adds $\lambda T_{\mathrm{KD}}^2 \mathrm{KL}(\sigma(\ell^0/T_{\mathrm{KD}}) \| \sigma(\ell'/T_{\mathrm{KD}}))$ over training patches sampled without labels. Here σ is softmax, ℓ^0 and ℓ' are frozen and corrected logits, and T_{KD}=2 is the distillation temperature. λ weights distillation, and the separate trust-region bound τ clips each logit residual to $[-\tau, \tau]$.

Pseudo-novel episodes. B5 withholds 30 seen classes per epoch and fits cross-entropy on the remaining seen-class patches. It also matches frozen probabilities normalized within the withheld class columns, using a KL penalty of weight 1 and temperature 2. Because the actual novel classes are outside this penalty, it does not constrain their probability relative to seen classes.

Masked-crop graph decoder. The decoder in Tables 2 and S4 uses the graph, embedding residual, calibration and prototype components of main-paper §4, with 768-dimensional region features, logit scale 100, and a 0.2 $\tanh$ embedding residual. It trains for 12 epochs with Adam (learning rate 10^{-3}, weight decay 10^{-4}). Its auxiliary unknown-region score is trained against ignored proposal labels and converted to a probability with sigmoid. Proposals above 0.5 are

excluded from painting, while ECE uses all labeled proposals. Novel vocabulary classes remain semantic labels. This decoder is distinct from the dense MLP.

Unseen-class parameters in per-class calibrators. We update only seen-class parameters. Global temperature T_{global} divides every logit by one positive scalar. Per-class temperature uses T_c for class c, while vector scaling applies $a_c \ell_c + b_c$ with scale a_c and bias b_c. In the masked-crop calibration runs (S1, S5 and Fig. S1), novel classes inherit the fitted global temperature: $T_c = T_{\text{global}}$, or $a_c = 1/T_{\text{global}}, b_c = 0$ for vector scaling. Dense-CLIP and VOC per-class temperature runs instead leave novel classes at $T_c = 1$. These conventions both avoid fitting novel-class parameters but use different reference scales.

Confidence and ECE. Masked-crop ECE is computed once per labeled proposal against its majority ground-truth label, over all proposals whose label is not ignored, with 15 equal-width confidence bins. It is computed before painting and does not depend on which overlapping proposal wins a pixel. Pixel IoU uses the highest-confidence covering proposal. Dense ECE uses labeled patches directly. Adaptive-bin and class-balanced dense ECE variants are recorded separately in the artifact.

Diffusion hyperparameters. α=0.4 and k=2 were fixed before novel-class evaluation. Table S2 reports the full weight/step sweep, without selecting on novel labels. In contrast, η denotes the residual interpolation scale in B4; it is unrelated to the diffusion weight.

Dense graph and learned affinity. Per image, our control connects each patch to its nearest neighbors by cosine similarity, forming a k-nearest-neighbor (kNN) graph (k_{nn}=8) on normalized CLIP patch features, excludes self-edges, clips negative similarities, symmetrizes the weights, and row-normalizes them (denominators clamped at 10^{-6}). With normalized adjacency matrix A, diffusion iterates $Q \leftarrow (1-\alpha)Q_0 + \alpha AQ$ for k steps. Here Q stacks patch probabilities and Q_0 holds the frozen probabilities, distinct from the text prototypes P in the main paper. The generic Laplacian drops the anchor term. B6 learns graph affinities within this anchored update. The cost record averages 576 nodes and 4608 directed neighbor entries per image. LPOSS uses a separate DINO graph with k_{nn}=400. The four propagation functions match the recorded upstream commit by SHA-256 checksums, and exact nearest-neighbor selection replaces the GPU search kernel.

Drift accounting. Drift follows Eq. (2) in the main paper, using corrected and frozen probability vectors on the same valid patch support. The 38 class-specific sweep settings include the frozen B4 endpoint and repeated unanchored endpoints: 32 distinct correction configurations remain after excluding the frozen endpoint and retaining one unanchored-head setting. The reported pointwise and family-bootstrap statistics use all 38 measured settings; excluding the frozen endpoint and repeated head configurations gives approximately $r = -0.88$. The fit in Fig. 4 uses those 38 settings, excluding the class-symmetric controls.

Independent rebuild. Reproduction deviations are computed by `scripts/repro_deviation.py` and stored in `results/extended/repro_deviation.json`. Compared dense rows cover the uncalibrated base, global and per-class temperatures, diffusion and Laplacian, agreeing within 0.05; temperatures are fitted, not parameter-free. Trained-decoder metrics differ by at most 0.85 (VOC novel ECE 16.7 versus 17.5), with ADE maximum 0.40. Trained-head AURCs are undefined in the rebuild and excluded. There is no independent rebuild record for the alignment/frontier, diagnostic, transfer or LPOSS experiments (Tables 5, 6, S7–S9). The masked-crop rebuild reproduces the qualitative pattern at 10.2 baseline mIoU on 1623 cacheable images. A repeated run on that rebuilt cache agrees within 1.2 for decoder-ablation metrics and within 0.02 for five-split calibration. Absolute results from independently generated SAM caches are not directly comparable.

M Claims and scope

We connect each central claim to its evidence and evaluated scope in Table S11. OVSS denotes open-vocabulary semantic segmentation. The collapse index C measures novel-class IoU relative to the frozen baseline, as defined in the main paper. Reading the evidence and scope together separates measured retention, the empirical drift association and the deployment implications.

Table S11: Claims, supporting evidence and evaluation scope. Protocols appear in Tables 1 and S0.

Claim	Evidence	Scope
Novel-class loss. Seen-only, class-specific, unanchored corrections reduce novel-class IoU, often to near zero.	Dense regime: 15.1→0.1 (Table 5). Masked crops: 9.4→1.1 on the original cache (Tables 2 and S4) and 9.5→2.5 on the independent rebuild (Supp. A). Per-class T and vector scaling behave alike (Table S1).	Five bases: masked crops, dense CLIP, LPOSS, DINO-proxy and VOC. ADE20K bases span 9–15.5 mIoU and VOC 75–82. Masked-crop accuracy uses seed 0, and the dense and calibration comparisons use five splits. CorrCLIP and CASS are untested.
Drift as a marker. Alignment drift tracks the loss within the class-specific family. Class-symmetric controls are exceptions.	Pointwise $r=-0.90$ (cluster-bootstrap 95% CI $[-0.96, -0.74]$) and family-level $r=-0.65$ ($[-0.95, -0.23]$, 13 families, Fig. 4). Global T (drift 13.4) and LPOSS (drift 50.6) keep novel-class IoU, as does B6, a learned affinity inside anchored diffusion ($C=1.04$).	An empirical association. Sweeps within one family count as correlated evidence. Drift serves as a monitoring and intervention axis, and no causal claim is made.
Anchoring. Distilling the head toward the frozen logits restores novel-class IoU and improves mIoU.	At $\lambda=30$, novel-class IoU is 15.3 ($C=1.01$) and mIoU rises from 14.5 to 15.8. Retention is monotone in λ, g, η and τ (Tables 5, S7 and S8, Fig. 3).	Dense regime, five splits. Anchored heads label fewer evaluation patches as novel (7–11% against 15.6%). IoU recovery coexists with a smaller predicted novel region.
Confounds. Class imbalance, objective, logit scale, supervision breadth and label purity do not explain the loss.	Each factor is varied in isolation. Novel-class IoU stays at or below 0.3 except at the narrowest supervision (30 seen classes, 3.3 ± 4.3, $C=0.22$). Only the trust region recovers it ($\tau=0.5$ gives 14.6, Table S8).	Adding seen classes lowers novel-class IoU further. Confounds we did not vary are outside the claim.
Calibration and controls. Among corrections that keep novel-class IoU, a global T calibrates best in domain. Diffusion and LPOSS serve as class-symmetric controls.	ECE 27.2→8.7 (dense) with novel-class IoU unchanged. The heads that lose the novel classes reach the lowest ECE and AURC (Table S6). The generic Laplacian matches anchored diffusion, and LPOSS is compared under a matched protocol.	Neither ECE nor AURC flags the loss, which only novel-class IoU reveals. We claim no gain over LPOSS or over trained OVSS systems.